CS5486
Intelligent Systems
Fall Semester, 2026

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Reference Books:

1) R. Rojas, *Neural Networks: A Systematic Introduction*, Springer, 1996.
2) C.-T. Lin and C.S. G. Lee, *Neural Fuzzy Systems*, Prentice Hall, 1996.
3) S. Haykin, *Neural Networks: A Comprehensive Foundation* (Third Ed.), Prentice-Hall, 2009.
4) R. Kruse, C. Borgelt, F. Klawonn, C. Moewes, M. Steinbrecher, and P. Held, *Computational Intelligence - A Methodological Introduction,* Springer, 2013.

Lecture and tutorial sessions:

Thursday 19:00-21:50
Room 4057, Bank of China Building

Course Outline:

Concepts, models, and methods of intelligent systems. Topics include neural networks, support vector machines, fuzzy logic, simulated annealing, genetic algorithms, and their applications

Grading Policy:

Midterm exam 40%, final exam 50%, project assignments: 10%

Course Objectives

This course provides students with

1. ability to understand basic concepts of intelligent systems,
2. ability to solve various modeling and computational problems using computational intelligence techniques,
3. ability to apply various computational intelligence approaches in scientific and engineering applications encompassing mechanical engineering, electrical engineering, and manufacturing engineering.

Teaching/Learning Expectations

This is to remind students that they should not skip classes casually without justification and should remain silent during lecture sessions (for the sake of others in the class and to respect the instructor's rights). This is to remind students that they should not skip classes casually without justification and should remain silent during lecture sessions (for the sake of others in the class and to respect the instructor's rights). Similarly, the course instructor/tutors should not skip or cancel classes/tutorials (cancellation or postponement of classes requires justification and official approval from the department/college office).

Academic Honesty

Students are expected to conform to the highest standards of honesty and integrity. Students are encouraged to discuss course material to foster the motivation of ideas and produce high-quality works. They may collaborate on the preliminary stages of individual homework assignments, but the final work must reflect their originality and individual effort. Plagiarism is considered a disciplinary offense that can result in reduced grades, failed subjects, and suspension from the university.